whether we've run headlong into a conflict with special relativity. A feature common to matter and energy is that when traveling from place to place they can transmit information. Photons traveling from a broadcast station to your radio carry information. Electrons traveling through Internet cables to your computer carry information. So, in any situation where something—even something unidentified—is purported to have traveled faster than light speed, a litmus test is to ask whether it has, or at least could have, transmitted information. If the answer is no, the standard reasoning goes, then nothing has exceeded light speed, and special relativity remains unchallenged. In practice, this is the test that physicists often employ in determining whether some subtle process has violated the laws of special relativity. (None has ever survived this test.) Let's apply it here.

Is there any way that, by measuring the spin of the left-moving and the right-moving photons about some given axis, we can send information from one to the other? The answer is no. Why? Well, the output found in either the left or the right detector is nothing but a *random* sequence of clockwise and counterclockwise results, since on any given run there is an equal probability of the particle to be found spinning one way or the other. In no way can we control or predict the outcome of any particular measurement. Thus, there is no message, there is no hidden code, there is no information whatsoever in either of these two random lists. The only interesting thing about the two lists is that they are identical—but that can't be discerned until the two lists are brought together and compared by some conventional, slower-than-light means (fax, e-mail, phone call, etc.). The standard argument thus concludes that although measuring the spin of one photon appears instantaneously to affect the other, no information is transmitted from one to the other, and the speed limit of special relativity remains in force. Physicists say that the spin results are correlated—since the lists are identical—but do not stand in a traditional cause-and-effect relationship because nothing travels between the two distant locations.

Entanglement and Special Relativity: The Contrarian View

Is that it? Is the potential conflict between the nonlocality of quantum mechanics and special relativity fully resolved? Well, probably. On the